

Mahdi Salehzadeh

Curriculum Vitae

Department of Mathematics, Simon Fraser University
8888 University Dr, Burnaby, BC. V5A 1S6 Canada
mahdi_salehzadeh@sfu.ca | msalehzadeh.github.io
Citizenship: Canada, Iran

EDUCATION

- Ph.D. Applied Mathematics**, Simon Fraser University, BC, Canada 2023 – current
Thesis: Models of Life-History Event Timing and Implications for Epidemiological, Ecological, and Evolutionary Dynamics
Advisor: Ailene MacPherson, John M. Stockie
- MSc. Applied Mathematics**, Simon Fraser University, BC, Canada 2021 – 2022
Thesis: Computational Study of 2D Jellyfish with the Immersed Boundary Method
Advisor: John M. Stockie
- BSc. Computer Science**, Amirkabir University of Technology, Tehran, Iran 2015 – 2019
Project: Cardinality Optimization
Advisor: S. A. Mirhassani

PUBLICATIONS

Theses

- M. Salehzadeh**. Computational study of 2d jellyfish with immersed boundary method. MSc thesis defended with distinction published at Simon Fraser University library.
Available at: <https://summit.sfu.ca/item/35829>

Peer-Reviewed

- M. Salehzadeh**, J.M. Stockie and A. MacPherson. An ecological model of masting reproduction matches empirical dynamics. *Theoretical Population Biology*, 160:62-69, 2024.
doi:<https://doi.org/10.1016/j.tpb.2024.10.002>

Under Review

- M. Salehzadeh**, J.M. Stockie and A. MacPherson. An ecological model of masting reproduction matches empirical dynamics. Under review in *Theoretical Ecology*.
bioRxiv doi:<https://doi.org/10.64898/2026.09.02.748907> [preprint]
- M. Salehzadeh**, J.M. Stockie and A. MacPherson. Cyclic pathogen epidemics favour the evolution of delayed germination. Under review in *Journal of Evolutionary Biology*.
bioRxiv doi:[NA](#) [preprint]
- M. Foisy, A. MacPherson, Y. McDonough, L.K. M'Gonigle, M. Maghrazi and **M. Salehzadeh**. A not so disturbing view of life history evolution. Under review in *American Naturalist*.

TEACHING

- Stochastic Modelling and Simulation**, MATH 348. Spring 2025, Spring 2026
Graduate Teaching Assistant. SFU.
- Numerical Analysis**, MACM 316. Fall 2025, Summer 2024, Fall 2023, Fall 2021
Graduate Teaching Assistant. SFU.
- Introduction to Biomathematics**, MATH 360. Fall 2023
Graduate Teaching Assistant. SFU.
- Mathematics in Sports**, MATH 303. Fall 2022, Fall 2024

Graduate Teaching Assistant. SFU.

Applied Calculus Workshop, ACW.

2021, 2022, 2023

Graduate Teaching Assistant. SFU.

Applied Linear Algebra, MATH 232.

Summer 2022

Graduate Teaching Assistant. SFU.

Complex Variables, MATH 322.

Fall 2024

Grader. SFU.

RESEARCH DISSEMINATION

Invited

Workshop: Biology and Modeling.

BaM!, Westfield, NC, US. (June 2026)

Workshop: Life-History Event Timing Drives Emergent Ecological Patterns.

BIRS, Banff, AB, Canada. (February 2026)

Conference: Evolution of Seed Masting (Or Not?).

CAIMS, Kingston, ON, Canada. (June 2024)

Contributed

Conference: Life-History Delays, Stochasticity, and the Emergence of Masting Reproduction.

CSEE, Toronto, ON, Canada. (May 2026)

Conference: Mathematical Models of Synchrony: From External Forcing to Stage-Structure.

SIAM, Montreal, QC, Canada. (July 2025)

Conference: Evolution of Seed Masting (Or Not?).

CSEE, Vancouver, BC, Canada. (May 2024)

Conference: Computational Study of 2D Jellyfish with the IBM.

CAIMS, Fredericton, NB, Canada. (June 2023)

Retreat: A Sequential Approach to Bark Beetle Outbreaks.

Eco-Evo, Squamish, BC, Canada. (October 2023)

Conference: A Sequential Approach to Bark Beetle Outbreaks.

SIAM PNW, Bellingham, WA, US. (October 2023)

HONORS AND RANKING

First Poster Presentation Award

2023

SIAM PNW, Bellingham, WA, US

MSc Thesis with Distinction

2022

Mathematics Department, SFU, BC, Canada

Top 5 Percent Rank in GPA in BSc.

2019

AmirKabir University of Technology, Tehran, Iran

Top 5 Percent Rank

2014

Iranian National Universities Entrance Exam (Mathematics and Physics branch)

Top 5 Percent Rank

2014

Iranian National Universities Entrance Exam (Foreign Languages)

FUNDING AND AWARDS

Travel & Minor Research Award, SFU, Summer 2026

1,000 CAD

Research Accelerator, Maud Menten Institute, Summer 2025

5,000 CAD

Travel & Minor Research Award, SFU, Summer 2025	1,200 CAD
Travel & Minor Research Award, SFU, Summer 2024	1,200 CAD
PhD Graduate Fellowship, SFU,	7,000 CAD/2Year
Travel & Minor Research Award, SFU, Fall 2023	900 CAD
First Place Poster Award, SIAM PNW, Fall 2023	100 USD
Travel & Minor Research Award, SFU, Spring 2023	900 CAD
Travel Fund, CAIMS, Spring 2023	1,200 CAD
MSc Graduate Fellowship, SFU, Summer 2022	7,000 CAD
Travel & Minor Research Award, SFU, Spring 2022	800 CAD
Travel & Minor Research Award, SFU, Summer 2021	800 CAD

PROFESSIONAL ACTIVITIES

Peer Reviews	Journal of Forestry Research
University Services	President, Mathematics Graduate Caucus, SFU, Summer 2024 - Summer 2025
	Vice-President, Mathematics Graduate Caucus, SFU, Spring 2024
	Member of the Central Council, Student Cultural Association, AUT, 2017
	Member of the Central Council, Student Environmental Association, AUT, 2017
Working Groups	PIMS-Maud Menton Institute, 2024
Social Services	First Aid Volunteer, Iranian Red Crescent Society, 2017

RESEARCH INTERESTS

Mathematical Biology, Epidemiological Disease Modeling, Computational Fluid Dynamics, Dynamical Systems, Nonlinear Dynamics, Numerical Analysis, Optimization

TECHNICAL STRENGTHS

Programing Languages	Python, MATLAB, , R, C, Java
Software and Tools	L ^A T _E X, Word, Excel, Powerpoint

LANGUAGES

Persian, English, Turkish, Azeri